Sagar Sinha

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Education

MS Computer Science Dec 2025
Arizona State University ***GPA**: 3.93/4.0*

- **Relevant Coursework**: Data Mining, Applied Regression Analysis, Natural Language Processing, Data Visualization, Distributed Database Systems

BS Computer Engineering May 2022
Pandit Deendayal Energy University ***GPA**: 9.02/10.0*

- **Relevant Coursework**: Data Structures, Algorithms, RDBMS, Data Warehousing, Statistics, Machine Learning

Professional Experience

AWS Cloud Developer Intern, Mindsparked Mar 2026 – Present

- **Designed and built event-driven serverless backends** on **AWS** (Lambda, Step Functions, API Gateway), owning features end-to-end from architectural design through production support and reducing operational overhead by 30%.
- **Architected scalable data persistence** across **Aurora PostgreSQL** and **DynamoDB**, implementing advanced indexing and GSI design that maintained **99.9% uptime** and sub-50ms query response times.
- **Automated cloud infrastructure and CI/CD pipelines** using **Terraform (IaC)**, integrating IAM least-privilege policies and CloudWatch monitoring to enforce auditable, repeatable deployments across environments.
- **Technical Skills:** AWS (Lambda, Aurora, DynamoDB, Step Functions, API Gateway), Terraform, PostgreSQL, Python, CI/CD, Git.

Software Engineering Externship (AI/ML), Extern Sep 2025 – Dec 2025

- **Owned end-to-end design and delivery** of a production-ready **RAG-based OCR engine** in **Python** for a **financial-services AI Mortgage Assistant**, automating extraction and indexing of 200+ page documents into a searchable vector store.
- **Optimized a hybrid retrieval system** using **LlamaIndex**, tuning chunking and embeddings to achieve **2s median latency** and a Mean Reciprocal Rank (**MRR**) of **0.98** across the production prototype.
- **Established operational excellence** for the ML pipeline through real-time logging and system observability, enabling production-grade visibility into document routing, model outputs, and failure modes.
- **Technical Skills:** Python (PyMuPDF, PaddleOCR), Vector Databases (LlamaIndex), NLP, Linux, Observability.

Data Analyst, Arizona State University Jul 2024 – Dec 2025

- **Architected a cloud-native ELT pipeline** on **AWS (S3, Redshift)** for 50K+ records, leveraging **Selenium/BeautifulSoup** scraping of authenticated, JS-rendered sources into a centralized data-persistence layer.
- **Streamlined data integrity through automated testing**, developing unit-test scripts that reduced manual QA intervention by **60%** and accelerated reporting cycles for university leadership stakeholders.
- **Transformed wellness data into 40+ actionable KPIs** through automated dashboards, identifying high-risk behavioral trends with **25% higher accuracy** than legacy methods.
- **Technical Skills:** Python, AWS (Redshift, S3), SQL, Data Modeling, Cloud Infrastructure, Selenium.

Data Science Researcher (ML), Accredian Apr 2023 – Jul 2023

- **Refactored legacy ETL pipelines** for 3,000+ users by migrating monolithic scripts into **modular Python components**, reducing technical debt by 40% and improving maintainability of the production environment.
- **Led Root Cause Analysis (RCA) on production bottlenecks**, resolving critical platform issues with a **90% success rate** to stabilize system performance during peak traffic.
- **Technical Skills:** Python (PyTorch, NumPy), System Optimization, ML Infrastructure, Production Support.

Relevant Projects

Distributed Banking System

- Engineered a distributed, **sharded MongoDB** cluster on **Docker**, sustaining **200K+ concurrent requests** with **sub-70ms read latency** across horizontally partitioned shard nodes.
- Implemented automated **replica-set health checks** and failover monitoring, maintaining $>$ 99**% successful requests** at peak concurrency.
- **Technical Skills:** HTML, CSS, JavaScript, Go, MongoDB, Docker, Bash.

Socratic CUDA Tutor

- Won **2nd Prize and a $2,000 scholarship** at the **ASU AI Spark Challenge** for building an AI-powered CUDA tutoring platform with guided conversations, persistent chat history, and module-based learning paths.
- Designed interactive learning flows across CUDA fundamentals, memory optimization, and kernel development, backed by a **RAG pipeline (Faiss)** indexing **30,000+ CUDA kernels** for context-aware, retrieval-grounded tutoring.
- Built an in-browser compile-run-debug workflow with real-time **GPU performance monitoring**, enabling learners to observe kernel execution and memory utilization with each code iteration.
- **Technical Skills**: React, JavaScript, Flask, Python, Supabase, Ollama, RAG, CUDA, C/C++, Docker, GPU Monitoring.

Reasoning Capabilities in Discrete Diffusion LLMs

- Built an **evaluation harness** for benchmarking **Discrete Diffusion (Llada-8B) vs. Autoregressive (LlaMA-8B)** models across **GSM8K and AIME2025** datasets; demonstrated Diffusion SFT superiority on math tasks (**67% vs. 56% accuracy**).
- Orchestrated **repairability experiments** (self-consistency, guided retry), revealing that test-time compute boosts Llada accuracy by **13.5%** with zero over-repair compared to AR models.
- **Technical Skills:** Python (Torch), Huggingface, LLM, Supervised Finetuning, Prompting, Evaluation Metrics.